Chhaya · System Architecture

A diabetes digital twin that shows its work — from one sensor wear to a patient's shadow

TEAM SYNAPSEX · IIT KANPUR
Happiest Health Digital Twin Challenge 2026

THE PROBLEM

A sensor lasts 14 days and costs ₹4,200. So a patient wears it once, gets a report, and is treated from that report for months. Nobody tells the doctor how old the report has become.

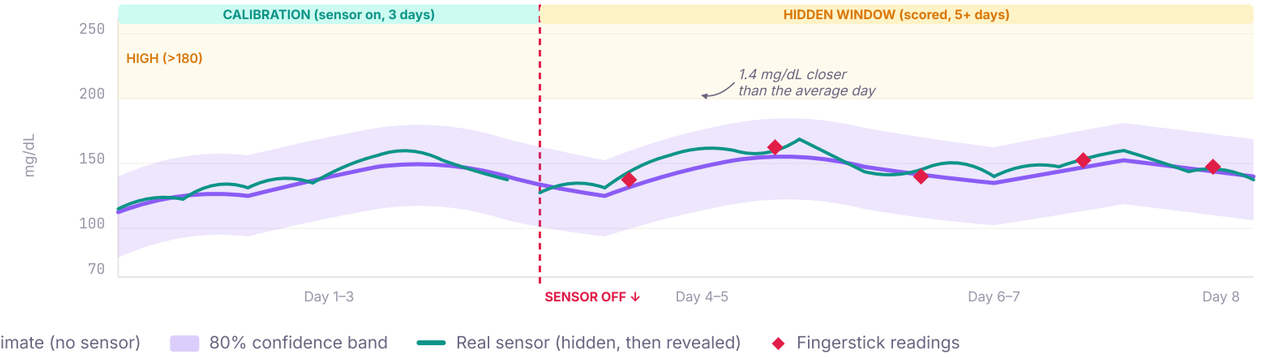
101M people with diabetes in India

₹50K per year for continuous CGM

14d one sensor wear, then nothing

THE CORE CONCEPT — "THE REVEAL"

Chhaya's estimate (purple with 80% band) is drawn **without the sensor**. Then the real sensor trace (teal) slides over it. The doctor sees where the shadow was right — and where it missed.



DATA INPUTS

CGM Sensor Wear — CALIBR.
3–14 days, 5-min intervals · 145 patients across 2 datasets

Health Record (EHR) — PRIOR
Fasting glucose, HbA1c, demographics, diagnoses, labs

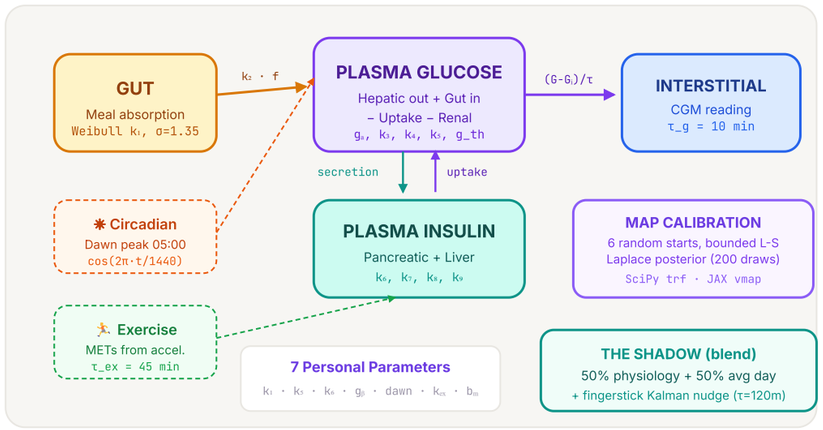
Meal Log — ALL
Timestamp + carbs with fat/protein/fibre slowing factor

Fingerstick Glucometer — POST
Sparse post-sensor readings, mapped to CGM scale

Wrist Accelerometer — CALIBR.
METs → exercise term on insulin-dependent uptake

🔒 **LEAKAGE BOUNDARY**
Sensor data never crosses the calibration split. Every feature builder is tested: change hidden sensor → check nothing else moves.

PHYSIOLOGICAL TWIN ENGINE (E-DES IN JAX)



INTEGRATOR
RK4, 1-min grid, jax.lax.scan

CALIBRATION
MAP fit + Laplace band, ~2s/patient

UNCERTAINTY
200-member ensemble via jax.vmap

STALENESS
CUSUM on fingerstick surprises

EVIDENCE SCORECARD — BARS WRITTEN BEFORE THE RUN

CLINICAL DASHBOARD — THREE SCREENS, TEN-SECOND READ

Built to be read by a doctor in a 4–5 minute consult. Runs offline from pre-computed JSON. **Nothing is fitted while it runs.** No dose advice. No alarm. An estimate is never drawn without its band and the word "estimated".

EVALUATION PIPELINE

1. **Split by Patient, Never by Row**
Half tunes settings. Half scored once. `--confirm` flag on committed code.

2. **Write Bars Before the Run**
Success criteria registered in `PREREGISTRATION.md` with dated amendments. No moving goalposts.

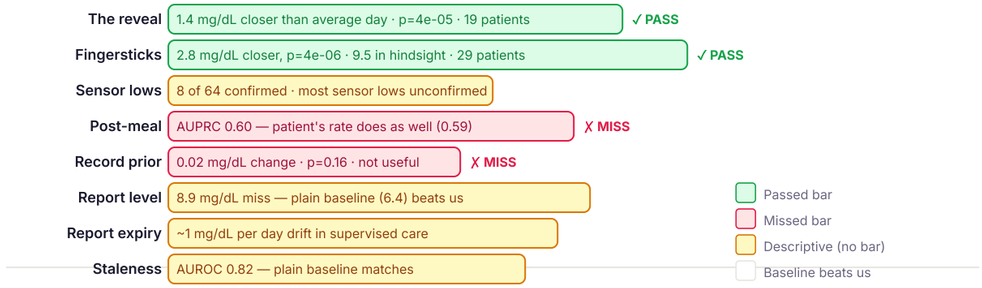
3. **Calibrate on k Days of Sensor**
MAP fit of 7 params. Record sets prior. Tested k=3, 5, 7.

4. **Simulate the Shadow**
50% ODE physiology + 50% patient's average day. Laplace ensemble → 80% band.

5. **Assimilate Fingersticks**
Kalman filter with fading deviation (τ=120min). RTS smoother for retrospective report.

6. **Detect Report Staleness**
CUSUM of standardised surprises. AUROC 0.82 for drift detection.

7. **Publish Misses With Passes**
Every result → JSON. Tests check each figure value against claim wording. Numbers and words cannot drift.



5	4	6	0
Bars passed	Bars missed (published)	Descriptive results	Leakage events

SCREEN 1

Clinic List

- Which patients have an old report
- Report age and staleness flag
- Pseudonymised names
- "Synthetic" labels

SCREEN 2

Patient Shadow

- Shadow first, sensor slides over
- Purple estimate + 80% band
- Teal/amber real sensor overlay
- What-if meal simulation

SCREEN 3

Evidence Wall

- Every result beside its bar
- Misses shown, not hidden
- Figures + regeneration cmd
- Limits on the screen

DESIGN PRINCIPLES — WHAT SEPARATES THIS FROM A MODEL VALIDATED ON ITS OWN SIMULATOR

No Leakage

Every feature builder tested: change the hidden sensor, check nothing else moves

Split by Patient

Development / held-out. Scored once. -- confirm on committed code

Bars Written First

Pre-registered criteria with dated amendments. No moving the goalposts

Misses Published

A failed fit is a row with an error, not a dropped patient. 4 missed bars on every page

One-Command Repro

Every result regenerated by a single command. Results committed, code committed

Synthetic Labelled

"Mrs. R." is simulated and labelled so on screen. No genetic markers in either dataset

`Python 3.12` `JAX` `SciPy` `NumPy` `Pandas` `pytest` `RK4 ODE` `Laplace Ensemble` `Kalman Filter` `CUSUM` `HTML/CSS/JS` `MIT License`

✗ Does not replace a sensor · ✗ No low-glucose estimate · ✗ No dose advice · ✗ No Indian data yet · ✗ No drug kinetics